

Pavan Reddy Thirumal Reddy

GenAI Developer | LLM Systems | RAG | Python | FastAPI | Scalable AI Infrastructure

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PROFESSIONAL SUMMARY STATEMENT

GenAI Developer with experience building LLM-powered applications, RAG systems, backend APIs, evaluation workflows, and scalable AI inference services. Skilled in Python, FastAPI, Azure OpenAI, LangChain, LlamaIndex concepts, FAISS, Pinecone, embeddings, prompt engineering, Docker, CI/CD, MLflow, logging, monitoring, and output validation. Experienced designing end-to-end GenAI architectures including ingestion, chunking, embedding pipelines, vector retrieval, API serving, evaluation, fallback handling, and production troubleshooting. Strong focus on improving response relevance, reducing hallucinations, optimizing latency, and delivering reliable GenAI applications across business and data-driven workflows.

Work Authorization: Authorized to work in the U.S.

Location: Open to remote, hybrid, and onsite roles across the U.S.

TECHNICAL SKILLS

- **Programming:** Python, SQL, JavaScript, C++
- **GenAI & LLMs:** Azure OpenAI, GPT-4, GPT-3.5, OpenAI APIs, Prompt Engineering, Context Optimization, LLM Evaluation
- **RAG Systems:** LangChain, LlamaIndex Concepts, FAISS, Pinecone, Embeddings, Chunking, Semantic Search, Retrieval Optimization
- **Backend & APIs:** FastAPI, Flask, REST APIs, Async Processing, API Integration, Backend Services, Microservices
- **AI Reliability:** Response Validation, Hallucination Reduction, Retry Logic, Fallback Handling, Guardrails, Human Review
- **ML Frameworks:** PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, NLP, Model Evaluation
- **Data Platforms:** BigQuery, MongoDB, SQL, NoSQL, Data Processing, Data Validation, ETL Pipelines
- **Cloud & MLOps:** Azure OpenAI, AWS Concepts, Google Cloud Platform, Docker, Kubernetes, GitHub, CI/CD, MLflow, Model Monitoring
- **Observability:** Logging, Metrics, Latency Tracking, Token Usage Monitoring, Root-Cause Analysis, Production Troubleshooting
- **Tools:** Git, GitHub, Linux/Unix, Jupyter Notebook, VS Code

PROFESSIONAL EXPERIENCE

Kentucky Energy & Environment Cabinet (EEC) — Data Analyst | June 2026 – Present **Frankfort, Kentucky**

- Maintains regulated environmental and operational data across ARM, SMIS, DOCTREE, and internal agency systems supporting state environmental programs
- Tracks datasets used for state and federal reporting, ensuring data quality, consistency, traceability, and audit-ready compliance records
- Prepares reports, forms, correspondence, and documentation while supporting stakeholder communication in a regulated government environment

Sodexo Live — Machine Learning Engineer Student Worker, Business AI & GenAI Automation | February 2025 – April 2026 Cincinnati, Ohio

- Supported business GenAI and automation initiatives for food service and live-event operations using Python, FastAPI, LLM APIs, RAG workflows, and data automation, reducing manual reporting and analysis effort by 40%
- Designed Retrieval-Augmented Generation pipelines using embeddings, FAISS, Pinecone, chunking, semantic search, and prompt workflows, improving response accuracy by 30%
- Reduced hallucinated outputs by 25% through retrieval optimization, prompt evaluation, response review, and human-in-the-loop validation workflows
- Developed asynchronous FastAPI services supporting concurrent LLM inference requests, retry logic, batching, and API error handling for reliable AI application workflows
- Optimized LLM response latency by approximately 20% through prompt compression, context selection, caching strategies, and backend performance improvements
- Built evaluation frameworks measuring retrieval quality, response relevance, latency, failure rates, token usage, and business KPIs to improve GenAI system reliability
- Implemented logging, monitoring, validation checks, and troubleshooting workflows to track AI outputs, diagnose failures, and support continuous improvement
- Containerized GenAI services using Docker and integrated CI/CD pipelines with GitHub and MLflow to improve release repeatability and deployment traceability
- Created technical documentation, API notes, prompt evaluation summaries, and setup guides to support maintainability and cross-functional adoption

Donato Technologies — AI / ML Research Assistant | Jun 2023 – Jul 2024 | Hybrid **Hyderabad, India**

- Developed NLP and transformer-based models using Python, PyTorch, TensorFlow, and Hugging Face for text classification, semantic understanding, and summarization workflows
- Built FastAPI-based AI inference APIs enabling real-time predictions, retrieval workflows, and scalable AI service integrations
- Implemented embedding-based semantic retrieval pipelines and vector search workflows supporting contextual question answering across document datasets
- Conducted model evaluation using accuracy, precision, recall, F1-score, and qualitative review methods to improve prediction reliability and response quality
- Communicated model performance, limitations, assumptions, and improvement opportunities to technical reviewers through clear documentation and structured findings

Cognifyz Technologies — Machine Learning Intern | Jan 2023 – May 2023 | Remote

- Developed machine learning models using Python and Scikit-learn, achieving approximately 85% classification accuracy across structured datasets
- Performed data preprocessing, feature engineering, exploratory data analysis, statistical analysis, and model evaluation for analytics and machine learning workflows

PROJECTS

RAG-Based Knowledge Assistant

- Built a Retrieval-Augmented Generation platform using Azure OpenAI, FAISS, Pinecone, embeddings, semantic search, and FastAPI to deliver contextual knowledge retrieval
- Designed ingestion, chunking, embedding, vector indexing, retrieval optimization, response generation, and API serving workflows for scalable GenAI applications

- Implemented retry logic, response validation, logging, and evaluation workflows to improve answer relevance, latency, failure handling, and system reliability

LLM-Powered Chatbot Application

- Developed an LLM-based chatbot with prompt orchestration, context-aware response generation, and backend APIs for real-time user interactions
- Built validation workflows to verify AI outputs, reduce incorrect responses, and improve reliability across common user queries
- Documented setup steps, API behavior, prompt patterns, and known limitations to support maintainability and technical onboarding

AI Workflow Automation System

- Designed AI-driven automation workflows using LLM APIs, REST APIs, structured datasets, and backend services to improve efficiency across repeatable tasks
- Integrated machine learning models with LLM workflows to support automated processing, decision support, and structured output generation

EDUCATION

University of Cincinnati, College of Engineering and Applied Science Master of Engineering, <i>Artificial Intelligence</i> CGPA: 3.6 • Coursework: Artificial Intelligence, Deep Learning, Cloud Computing, Intelligent Systems, Reinforcement Learning, NLP	Cincinnati, Ohio April 2026
Vardhaman College Of Engineering Bachelor of Technology, <i>Computer Science (Artificial Intelligence & Machine Learning)</i> • Coursework: Machine Learning, Web Development, Human Computer Interaction, Software Engineering	Hyderabad, India May 2024

CERTIFICATIONS & ACHIEVEMENTS

- Data Science with Python
- RPA Developer Foundation
- Cybersecurity Basics & Network Support
- Graduate Incentive Award - University of Cincinnati